Fiscal Year:	FY 2014	**Task Last Updated:**	FY 11/13/2014
PI Name:	Seidler, Rachael D. Ph.D.		
Project Title:	Spaceflight Effects on Neurocognitive Performance: Extent, Longevity, and Neural Bases		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rachaelseidler@ufl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	352-294-1722
Organization Name:	University of Florida		
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PI Web Page:			
City:	Gainesville	**State:**	FL
Zip Code:	32611-8205	**Congressional District:**	3
Comments:	NOTE: PI moved to University of Florida in July 2017; previous affiliation was University of Michigan.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Crew Health NNJ10ZSA003N
Start Date:	08/26/2011	**End Date:**	07/13/2017
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Loerch, Linda	**Contact Phone:**	
Contact Email:	linda.loerch-1@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 7/13/2017 due to transfer to PI's new institution (Ed., 1/11/2018) NOTE: Extended to 3/31/2019 per NSSC information and PI (Ed., 8/20/2012)		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Bloomberg, Jacob (NASA Johnson Space Center) Mulavara, Ajitkumar (Universities Space Research Association) Wood, Scott (NASA Johnson Space Center)		
Grant/Contract No.:	NNX11AR02G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NRA NNJ10ZSA003N requested proposals to assess changes in elemental neurocognitive functions such as perception, motor control, memory, attention, language, executive function, and emotional processing following long duration spaceflight using both behavioral assessments and monitoring technologies such as fMRI. In response to this call, we propose to perform structural and functional MR brain imaging to identify the relationship between changes in crewmember neurocognitive function and neural structural alterations following a six month International Space Station mission. Our central hypothesis is that measures of brain structure, function, and network integrity will change from pre to post flight in crewmembers (Aim 1). Moreover, we predict that these changes will correlate with indices of cognitive, sensory, and motor function in a neuroanatomically selective fashion (Aim 2). Our interdisciplinary approach utilizes cutting edge neuroimaging techniques and a broad ranging battery of sensory, motor, and cognitive assessments that will be conducted pre flight, during flight, and post flight to investigate neuroplastic and maladaptive brain changes in crewmembers following long duration spaceflight. Success in this endeavor would 1) result in identification of the underlying neural mechanisms and operational risks of spaceflight-induced changes in behavior, and 2) identify whether a return to normative behavioral function following re-adaptation to Earth's gravitational environment is associated with a restitution of brain structure and function or instead is supported by substitution with compensatory brain processes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this project will have relevance not only to understanding the effects of spaceflight on the human brain and behavior, but also for delineating the capacity of the brain to remodel in response to adaptive stimuli. As such, the results should prove informative for understanding the neural mechanisms associated with adaptive behavioral change and the rehabilitation of these changes during recovery periods.
Task Progress:	To date, we have collected preflight data of one astronaut. Data inspection revealed that the quality of the obtained MRI scans was good. In addition to the structural T1-weighted imaging data and the diffusion tensor imaging data, we added a state of the art high-resolution T2-weighted sequence to our neuroimaging protocol. Intracranial fluid shifts and flattening of the back of the eyeball have already been associated with microgravity, which is why the addition of T2-weighted imaging could be helpful in unraveling additional effects of spaceflight on the central nervous system. Discussion: We are making excellent progress. Given our ability to leverage data collection from our ongoing NSBRI-funded bed rest study, we have extensively refined our data collection and analysis approaches. We have obtained all of our necessary ethical, hardware, and software approvals, and have our first flight study subject enrolled. Both ground based normative control testing and testing of bed rest subjects will be completed by the end of the year. Estimated total numbers of completed participants for bed rest subjects will be 15 and 20 for the normative control subjects.
Bibliography Type:	Description: (Last Updated: 03/12/2026)
Articles in Peer-reviewed Journals	Koppelmans V, Erdeniz B, De Dios YE, Wood SJ, Reuter-Lorenz PA, Kofman I, Bloomberg JJ, Mulavara AP, Seidler RD. "Study protocol to examine the effects of spaceflight and a spaceflight analog on neurocognitive performance: extent, longevity, and neural bases." BMC Neurol. 2013 Dec 18;13:205. http://dx.doi.org/10.1186/1471-2377-13-205 ; PubMed PMID: 24350728; PubMed Central PMCID: PMC3878338 , Dec-2013